

7.5.2 External power supply inputs: VIN and E5V

The external power sources VIN and E5V are summarized in [Table 7](#). When the board is power supplied by VIN or E5V, the jumper configuration must be the following:

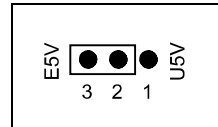
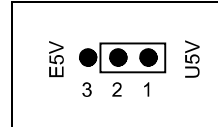
- Jumper on JP5 pin 2 and pin 3
- Jumper removed on JP1

Table 7. External power sources

Input power name	Connectors pins	Voltage range	Max current	Limitation
VIN	CN6 pin 8 CN7 pin 24	7 V to 12 V	800 mA	From 7 V to 12 V only and input current capability is linked to input voltage: 800 mA input current when $V_{in} = 7\text{ V}$ 450 mA input current when $7\text{ V} < V_{in} \leq 9\text{ V}$ 250 mA input current when $9\text{ V} < V_{in} \leq 12\text{ V}$
E5V	CN7 pin 6	4.75 V to 5.25 V	500 mA	-

Table 8. Power-related jumper

Jumper	Description
JP5	U5V (ST-LINK VBUS) is used as a power source when JP5 is set as shown below (Default setting)
	VIN or E5V is used as a power source when JP5 is set as shown below.



Using VIN or E5V as an external power supply

VIN or E5V can be used as an external power supply in case the current consumption of the STM32 Nucleo and extensions boards exceeds the allowed current on USB. In this condition, it is still possible to use the USB for communication, programming, or debugging only, but it is mandatory to power supply the board first using VIN or E5V and then connect the USB cable to the PC. Proceeding this way ensures that the enumeration occurs thanks to the external power source.

The following power sequence procedure must be respected:

1. Connect the jumper between pin 2 and pin 3 of JP5
2. Check that JP1 is removed
3. Connect the external power source to VIN or E5V
4. Power on the external power supply $7\text{ V} < \text{VIN} < 12\text{ V}$ to VIN, or 5 V for E5V
5. Check that LD3 is turned ON
6. Connect the PC to USB connector CN1

If this order is not respected, the board might be supplied by VBUS first then by VIN or E5V, and the following risks might be encountered:

1. If more than 300 mA current is needed by the board, the PC might be damaged or the current supply can be limited by the PC. As a consequence, the board is not powered correctly.
2. 300 mA is requested at enumeration (since JP1 must be OFF) so there is a risk that the request is rejected and the enumeration does not succeed if the PC cannot provide such current. Consequently, the board is not power supplied (LED LD3 remains OFF).

7.5.3 External power supply input: +3.3V

It can be of interest to use the +3.3V (CN6 pin 4 or CN7 pin 12 and pin 16) directly as power input for instance in case the 3.3V is provided by an extension board. When the STM32 Nucleo board is power supplied by +3.3V, the ST-LINK is not powered, thus the programming and debug features are unavailable. The +3.3V external power source is summarized in [Table 9](#).

Table 9. +3.3 V external power source

Input power name	Connectors pins	Voltage range	Limitation
+3.3V	CN6 pin 4 CN7 pin 12 and pin 16	3 V to 3.6 V	Used when the ST-LINK part of the PCB is cut or SB2 and SB12 OFF

Two different configurations are possible when using +3.3V to power the board:

- ST-LINK is removed (PCB cut) or
- SB2 (3.3V regulator) and SB12 (NRST) are OFF.

7.5.4 External power supply output

When powered by USB, VIN, or E5V, the +5V (CN6 pin 5 or CN7 pin 18) can be used as an output power supply for an ARDUINO® shield or an extension board. In this case, the maximum current of the power source specified in [Table 7](#) must be respected.

The +3.3V (CN6 pin 4 or CN7 pin 12 and 16) can be used also as power supply output. The current is limited by the maximum current capability of the regulator U4 (500 mA max).